



Discrete Mathematics Notes 3: Set Theory

Author: Qinxiang Cao

Institute: Shanghai Jiao Tong University

Date: June 18, 2026

Contents

Chapter 1 Naive Set Theory	B
1.1 Introduction	B
1.2 Binary Relations	C
1.3 Equivalence Relation	D
1.4 Partitions	E
1.5 Closures	F
1.6 Functions	G
1.7 Bernstein's Theorem	I
1.8 Equinumerosity	J
1.9 Countable Sets	L
1.10 Uncountable Sets	L

Chapter 1 Naive Set Theory

1.1 Introduction

Definition 1.1

- $x \in y$
- $x = y$ iff $\forall z(z \in x \leftrightarrow z \in y)$
- $x \subseteq y$ iff $\forall z(z \in x \rightarrow z \in y)$
- $x \cup y$: $z \in x \cup y$ iff $z \in x$ or $z \in y$
- $x \cap y$: $z \in x \cap y$ iff $z \in x$ and $z \in y$
- $x \setminus y$: $z \in x \setminus y$ iff $z \in x$ and $z \notin y$
- Empty set $\emptyset$: for any x , $x \notin \emptyset$
- Power set $\mathcal{P}(x)$: $y \in \mathcal{P}(x)$ iff $y \subseteq x$
- General union $\bigcup x$: $y \in \bigcup x$ iff $\exists z$ such that $z \in x$ and $y \in z$
- Cartesian product $x \times y$: $\{(u, v) \mid u \in x \text{ and } v \in y\}$



Example 1.1

1. If $A \subseteq B$, then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.
2. If $A \subseteq B$, then $\bigcup A \subseteq \bigcup B$.
3. $A = \bigcup(\mathcal{P}(A))$.

Proof (1) Let $X \in \mathcal{P}(A)$. Then $X \subseteq A$. Since $A \subseteq B$, we have $X \subseteq B$, so $X \in \mathcal{P}(B)$.

Hence $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

(2) Let $x \in \bigcup A$. Then there exists $Y \in A$ such that $x \in Y$. $A \subseteq B$ implies $Y \in B$, so $x \in \bigcup B$.

(3) “ $\subseteq$ ”: Let $x \in A$. Then $\{x\} \in \mathcal{P}(A)$, and $x \in \{x\}$, so $x \in \bigcup \mathcal{P}(A)$. Thus $A \subseteq \bigcup \mathcal{P}(A)$.

“ $\supseteq$ ”: Let $x \in \bigcup \mathcal{P}(A)$. By definition there exists $\gamma \in \mathcal{P}(A)$ such that $x \in \gamma$.

Since $\gamma \in \mathcal{P}(A)$, $\gamma \subseteq A$, hence $x \in A$.

Example 1.2

1. $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$
2. $A \times \bigcup B = \bigcup \{A \times X \mid X \in B\}$

Proof (1): “ $\Rightarrow$ ”: Let $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$. Then $X \in \mathcal{P}(A)$ and $X \in \mathcal{P}(B)$, so $X \subseteq A$ and $X \subseteq B$.

For any $a \in X$, we have $a \in A$ and $a \in B$, hence $a \in A \cap B$. Therefore $X \subseteq A \cap B$, so $X \in \mathcal{P}(A \cap B)$.

“ $\Leftarrow$ ”: Let $X \in \mathcal{P}(A \cap B)$. Then $X \subseteq A \cap B$, so obviously $X \subseteq A$ and $X \subseteq B$. Thus $X \in \mathcal{P}(A)$ and $X \in \mathcal{P}(B)$, so $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$.

(2): “ $\Rightarrow$ ”: Let $(a, x) \in A \times (\bigcup B)$. Then $x \in \bigcup B$, so there exists $X \in B$ such that $x \in X$.

Hence $(a, x) \in A \times X$, and $A \times X \subseteq \bigcup \{A \times X \mid X \in B\}$. Thus $(a, x) \in \bigcup \{A \times X \mid X \in B\}$.

“ $\Leftarrow$ ”: Let $m \in \bigcup \{A \times X \mid X \in B\}$. Then there exists $X \in B$ such that $m \in A \times X$.

Write $m = (a, x)$ with $a \in A$ and $x \in X$. Since $X \in B$, $x \in \bigcup B$. Therefore $(a, x) \in A \times (\bigcup B)$.

1.2 Binary Relations

Definition 1.2

Subsets of $A \times B$ are binary relations from A to B . Write aRb for $(a, b) \in R$.



Definition 1.3

R is a binary relation on A if $R \subseteq A \times A$.

- **reflexive:** $\forall a \in A, aRa$.
- **symmetric:** $\forall a, b \in A$, if aRb then bRa .
- **antisymmetric:** $\forall a, b \in A$, if aRb and bRa then $a = b$.
- **transitive:** $\forall a, b, c \in A$, if aRb and bRc then aRc .



Definition 1.4

- **Inverse:** $R \subseteq A \times B, R^{-1} = \{(x, y) \in B \times A \mid yRx\}$.
- **Composition:** $R \subseteq A \times B, S \subseteq B \times C$. Then

$$S \circ R = \{(x, z) \in A \times C \mid \exists y (xRy \wedge ySz)\}.$$



Theorem 1.1

$$(R^{-1})^{-1} = R.$$



Proof For any $(a, b) \in (R^{-1})^{-1}$, we have $(b, a) \in R^{-1} \iff (a, b) \in R$.

Theorem 1.2

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}.$$



Proof For any $(a, c) \in (S \circ R)^{-1}$, we have $(c, a) \in S \circ R$, so $\exists b$ such that $(c, b) \in R$ and $(b, a) \in S$.

Then $(b, c) \in R^{-1}$ and $(a, b) \in S^{-1}$, hence $(a, c) \in R^{-1} \circ S^{-1}$.

Theorem 1.3

$$R \circ (S \circ T) = (R \circ S) \circ T.$$



Proof $(a, d) \in R \circ (S \circ T)$ iff $\exists c$ such that $(a, c) \in S \circ T$ and $(c, d) \in R$

iff $\exists b, c$ such that $(a, b) \in T, (b, c) \in S, (c, d) \in R$.

Theorem 1.4

$$R \circ (S \cup T) = (R \circ S) \cup (R \circ T).$$



Proof “ $\implies$ ”: if $(a, c) \in R \circ (S \cup T)$ then there exists b such that $(a, b) \in S \cup T$ and $(b, c) \in R$.

So aSb or aTb .

Hence $(a, c) \in R \circ S$ or $(a, c) \in R \circ T$, i.e. $(a, c) \in (R \circ S) \cup (R \circ T)$.

“ $\impliedby$ ”: if $(a, c) \in (R \circ S) \cup (R \circ T)$ then $(a, c) \in R \circ S$ or $(a, c) \in R \circ T$.

Case 1: $(a, c) \in R \circ S$.

Then there exists b such that $(a, b) \in S$ and $(b, c) \in R$.

Obviously $(a, b) \in S \cup T$, so $(a, c) \in R \circ (S \cup T)$.

Case 2: $(a, c) \in R \circ T$.

Similarly, there exists b with $(a, b) \in T$ and $(b, c) \in R$, thus $(a, b) \in S \cup T$ and $(a, c) \in R \circ (S \cup T)$.

Therefore $(a, c) \in R \circ (S \cup T)$.

Theorem 1.5

Let $R \subseteq A \times A$.

- R is reflexive on A iff $I_A = \{(a, a) \mid a \in A\} \subseteq R$.
- R is symmetric iff $R = R^{-1}$.
- R is antisymmetric iff $R \cap R^{-1} \subseteq I_A$.
- R is transitive iff $R \circ R \subseteq R$.



1.3 Equivalence Relation

Definition 1.5 (equivalence relation)

$R \subseteq A \times A$ is an **equivalence relation** on A if R is reflexive, symmetric and transitive.



Definition 1.6 (Equivalence class)

R is an equivalence relation on A , $a \in A$.

$$[a]_R = \{b \in A \mid aRb\}.$$



Theorem 1.6

aRb iff $[a]_R = [b]_R$.



Proof “ $\Rightarrow$ ”: If aRb , take any $c \in [a]_R$, i.e., aRc . Then by symmetry bRa , and by transitivity bRc , so $c \in [b]_R$. Similarly any $c \in [b]_R$ implies $c \in [a]_R$.

“ $\Leftarrow$ ”: Assume $[a]_R = [b]_R$. Since bRb , $b \in [b]_R$, so $b \in [a]_R$, thus aRb .

Theorem 1.7

aRb iff $[a]_R \cap [b]_R \neq \emptyset$.



Proof “ $\Rightarrow$ ”: Obvious.

“ $\Leftarrow$ ”: Assume $c \in [a]_R \cap [b]_R$. Then aRc and bRc .

By symmetry cRb , then transitivity gives aRb .

Example 1.3 Prove or disprove:

- If R_1 and R_2 are equivalence relations on A , then $R_1 \cup R_2$ is also an equivalence relation.
- If R_1 and R_2 are equivalence relations on A , then $R_1 \cap R_2$ is also an equivalence relation.

Proof (1) False. Counterexample: $A = \{a, b, c\}$,

$$R_1 = \{(a, a), (b, b), (c, c), (a, b), (b, a)\},$$

$$R_2 = \{(a, a), (b, b), (c, c), (b, c), (c, b)\}.$$

Then $(a, b) \in R_1$, $(b, c) \in R_2$, but $(a, c) \notin R_1 \cup R_2$, so $R_1 \cup R_2$ is not transitive.

(2) True.

Reflexive: $\forall a \in A, (a, a) \in R_1, (a, a) \in R_2$, then $(a, a) \in R_1 \cap R_2$.

Symmetric: If $(a, b) \in R_1 \cap R_2$, then $(a, b) \in R_1$ and $(a, b) \in R_2$. Since both are symmetric, $(b, a) \in R_1$ and $(b, a) \in R_2$, so $(b, a) \in R_1 \cap R_2$.

Transitive: If $(a, b) \in R_1 \cap R_2$ and $(b, c) \in R_1 \cap R_2$, then $(a, b), (b, c) \in R_1$ and $(a, b), (b, c) \in R_2$. Since both are transitive, $(a, c) \in R_1$ and $(a, c) \in R_2$, so $(a, c) \in R_1 \cap R_2$.

1.4 Partitions

Definition 1.7 (partition)

A **partition** of a set A is a collection of disjoint nonempty subsets of A that have A as their union.

Formally, $P \subseteq \mathcal{P}(A)$ is a partition of A if:

- $\forall B \in P, B \neq \emptyset$;
- $\forall B, B' \in P$, if $B \neq B'$ then $B \cap B' = \emptyset$;
- $\bigcup P = A$.



Theorem 1.8

If R is an equivalence relation on A , then $\{[a]_R \mid a \in A\}$ is a partition on A .



Proof (1) $\forall a \in A, aRa$, so $a \in [a]_R$, hence $[a]_R \neq \emptyset$.

(2) $\forall a, b \in A, aRb$ iff $[a]_R = [b]_R$ iff $[a]_R \cap [b]_R \neq \emptyset$.

(3) $\forall a \in A, a \in [a]_R$, so $[a]_R \in P$ implies $a \in \bigcup P$.

Theorem 1.9

Suppose P is a partition on A . Then $R = \{(a, b) \in A \times A \mid \exists B \in P \text{ such that } a \in B \text{ and } b \in B\}$ is an equivalence relation on A and $P = \{[a]_R \mid a \in A\}$.



Proof Reflexive: $\forall a \in A, \exists B \in P$ with $a \in B$, so aRa .

Symmetric: If aRb , then $\exists B \in P$ with $a \in B$ and $b \in B$, so bRa .

Transitive: Suppose aRb and bRc . Then $\exists B_1, B_2 \in P$ such that $a, b \in B_1$ and $b, c \in B_2$. Since $b \in B_1 \cap B_2$ and P is a partition, $B_1 = B_2$. Hence $a, c \in B_1$, so aRc .

Now show $P = \{[a]_R \mid a \in A\}$.

“ $\subseteq$ ”: Take $B \in P$, nonempty, pick $a \in B$. We prove $B = [a]_R$.

If $b \in B$, then aRb , so $b \in [a]_R$. Conversely, if $b \in [a]_R$, then aRb , so there exists $B' \in P$ with $a, b \in B'$. Since $a \in B \cap B'$ and P is a partition, $B = B'$, so $b \in B$.

“ $\supseteq$ ”: For any $a \in A$, there exists $B \in P$ with $a \in B$, and as above $B = [a]_R$. Thus $[a]_R \in P$.

Definition 1.8 (refinement)

A partition P_1 is called a **refinement** of partition P_2 if every set in P_1 is a subset of some set in P_2 .



Theorem 1.10

Let R_1, R_2 be equivalence relations on A with corresponding partitions P_1, P_2 . Then $R_1 \subseteq R_2$ iff P_1 is a refinement of P_2 .



Proof ($\Rightarrow$) Suppose $R_1 \subseteq R_2$. For any $a \in A$, we show $[a]_{R_1} \subseteq [a]_{R_2}$. If bR_1a , then bR_2a , so $b \in [a]_{R_2}$. Hence P_1 refines P_2 .

($\Leftarrow$) Suppose P_1 refines P_2 . Fix $a \in A$. There exists b such that $[a]_{R_1} \subseteq [b]_{R_2}$. Since $a \in [a]_{R_1}$, we have $a \in [b]_{R_2}$, so bR_2a and thus $[a]_{R_2} = [b]_{R_2}$. Hence $[a]_{R_1} \subseteq [a]_{R_2}$. If bR_1a , then $b \in [a]_{R_1} \subseteq [a]_{R_2}$, so bR_2a . Therefore $R_1 \subseteq R_2$.

1.5 Closures

Definition 1.9 (Transitive Closure)

T is the transitive closure of $R \subseteq A \times A$ if:

1. $R \subseteq T$,
2. T is transitive,
3. For any T' with $R \subseteq T'$ and T' transitive, we have $T \subseteq T'$.



Definition 1.10

Let $R^1 = R$, $R^{n+1} = R^n \circ R$, and $R^+ = \bigcup_{n=1}^{\infty} R^n$.



Lemma 1.1

If $R_1 \subseteq R'_1$ and $R_2 \subseteq R'_2$, then $R_1 \circ R_2 \subseteq R'_1 \circ R'_2$. (Obvious)



Lemma 1.2

If $R^n \subseteq T$ for all $n \in \mathbb{Z}^+$, then $\bigcup_{n=1}^{\infty} R^n \subseteq T$. (Obvious)



Lemma 1.3

$R \subseteq R^+$. (Obvious)



Lemma 1.4

$R^n \circ R^m = R^{n+m}$.



Proof [Proof by induction on m] (1) $m = 1$: $R^{n+1} = R^n \circ R$ by definition.

(2) $m = k + 1$: $R^n \circ R^{k+1} = R^n \circ (R^k \circ R) = (R^n \circ R^k) \circ R = R^{n+k} \circ R = R^{n+k+1}$.

Lemma 1.5

R^+ is transitive.



Proof Suppose aR^+b and bR^+c . Then $\exists n, m$ such that $aR^n b$ and $bR^m c$.

Then $(a, c) \in R^n \circ R^m = R^{n+m} \subseteq R^+$.

Lemma 1.6

If $R \subseteq S$ and S is transitive, then $R^+ \subseteq S$.



Proof It suffices to show $\forall n \in \mathbb{Z}^+, R^n \subseteq S$.

Base $n = 1$: $R \subseteq S$.

Inductive step: $R^{k+1} = R^k \circ R \subseteq S \circ S \subseteq S$ by transitivity of S .

Definition 1.11 (Reflexive Transitive Closure)

T is the reflexive transitive closure of R if:

1. $R \subseteq T$,
2. T is transitive and reflexive,
3. For any T' with $R \subseteq T'$ and T' transitive and reflexive, $T \subseteq T'$.

Define $R^* = \bigcup_{n=0}^{\infty} R^n$, where $R^0 = I_A = \{(a, a) \mid a \in A\}$.

**Theorem 1.11**

1. If $R \subseteq A \times A$ is symmetric, then R^+ is symmetric.
2. If $R, S \subseteq A \times A$ and $R \circ S \subseteq S$, then $R^+ \circ S \subseteq S$.
3. $(R^+)^+ = R^+$.



Proof (1) Suppose aR^+b , then $\exists n$ with $aR^n b$. By induction on n , using symmetry of R , we get $bR^n a$, so bR^+a .

$$(2) R^+ \circ S = (\bigcup_{n=1}^{\infty} R^n) \circ S = \bigcup_{n=1}^{\infty} (R^n \circ S).$$

Show $R^n \circ S \subseteq S$ by induction:

$n = 1$: $R \circ S \subseteq S$ by hypothesis.

Assume $R^k \circ S \subseteq S$, then $R^{k+1} \circ S = (R^k \circ R) \circ S = R^k \circ (R \circ S) \subseteq R^k \circ S \subseteq S$.

(3) Use the first definition.

1.6 Functions

Definition 1.12 (Function)

F is a **function** from A to B (denoted $F : A \rightarrow B$) iff:

1. $F \subseteq A \times B$,
2. $\forall a \in A, \exists b \in B$ such that aFb ,
3. If aFb_1 and aFb_2 , then $b_1 = b_2$.



Example 1.4 $F : \mathbb{Z} \rightarrow \mathbb{Z}, F(0) > 0$ means:

- $\exists a \in \mathbb{Z}$ with $(0, a) \in F$ and $a > 0$
- $\forall a \in \mathbb{Z}$, if $(0, a) \in F$ then $a > 0$

Theorem 1.12

If $F, G : A \rightarrow B$, then $F = G$ iff $\forall x \in A, F(x) = G(x)$.



Proof “ $\Rightarrow$ ”: Obvious.

“ $\Leftarrow$ ”: Suppose $(x, y) \in F$. Then $y = F(x) = G(x)$, so $(x, y) \in G$.

Hence $F \subseteq G$, similarly $G \subseteq F$.

Theorem 1.13

Suppose $F : B \rightarrow C$, $G : A \rightarrow B$.

1. $F \circ G : A \rightarrow C$.
2. $\forall x \in A, (F \circ G)(x) = F(G(x))$.



Proof (1) $F \circ G \subseteq A \times C$.

For any $x \in A$, $\exists b \in B$ with xGb , and $\exists c \in C$ with bFc , so $(x, c) \in F \circ G$.

Then prove its uniqueness: if $(x, c_1), (x, c_2) \in F \circ G$, then $\exists b_1, b_2$ such that $(x, b_1) \in G$, $(b_1, c_1) \in F$, $(x, b_2) \in G$, $(b_2, c_2) \in F$.

Since G is a function, $b_1 = b_2$, then $c_1 = c_2$ because F is a function.

(2) To prove $\forall x \in A, \forall b \in B, \forall c \in C$, if xGb and bFc , then $\exists d \in C$ such that $x(F \circ G)d$ and $d = c$.

Let $d = c$, then $(x, d) \in F \circ G$ iff $\exists b$ such that xGb and bFd .

Definition 1.13

- $F : A \rightarrow B$ is an **injection** (one-to-one) if $\forall x, y \in A, F(x) = F(y) \Rightarrow x = y$.
- F is a **surjection** (onto) if $\forall y \in B, \exists x \in A$ such that $F(x) = y$.
- F is a **bijection** if it is both injective and surjective.

**Theorem 1.14**

If $F : B \rightarrow C$ and $G : A \rightarrow B$ are both injections, then $F \circ G$ is an injection.



Proof If $(F \circ G)(x) = (F \circ G)(y)$, then $F(G(x)) = F(G(y))$. Since F is injective, $G(x) = G(y)$. Since G is injective, $x = y$.

Theorem 1.15

If F and G are surjections, then $F \circ G$ is a surjection.



Proof Let $c \in C$. Since F is surjective, $\exists b \in B$ with $F(b) = c$. Since G is surjective, $\exists a \in A$ with $G(a) = b$. Then $(F \circ G)(a) = F(G(a)) = F(b) = c$.

Theorem 1.16

If $F \circ G$ is an injection, then G is an injection. If $F \circ G$ is a surjection, then F is a surjection.



Proof (Injection part) Suppose $G(x) = G(y)$. Then $F(G(x)) = F(G(y))$, i.e., $(F \circ G)(x) = (F \circ G)(y)$. Since $F \circ G$ is injective, $x = y$.

(Surjection part) For any $c \in C$, since $F \circ G$ is surjective, $\exists a \in A$ such that $(F \circ G)(a) = c$. Then $F(G(a)) = c$, so c is image of $b = G(a)$ under F , hence F is surjective.

Theorem 1.17

If F is a bijection from A to B , then $F^{-1} \subseteq B \times A$ is also a bijection.



1.7 Bernstein's Theorem

Theorem 1.18 (Bernstein)

Suppose $f : A \rightarrow B$ and $g : B \rightarrow A$ are both injections. Then there exists a bijection between A and B . 

Proof Define $C_0 = \{a \in A \mid \forall b \in B, g(b) \neq a\}$ (elements of A not in the image of g).

$$D_0 = \{f(a) \mid a \in C_0\}$$

$$C_1 = \{a \in A \setminus C_0 \mid \forall b \in B \setminus D_0, g(b) \neq a\}$$

We can prove $C_1 = \{g(b) \mid b \in D_0\}$:

$a \in C_1$ iff $a \notin C_0$ and $\forall b \in B \setminus D_0, g(b) \neq a$.

$a \notin C_0$ means $\exists b \in B, g(b) = a$.

Thus $\exists b \in D_0, g(b) = a$.

Then

$$D_1 = \{f(a) \mid a \in C_1\}$$

Inductively,

$$C_{n+1} = \{a \in A \setminus \bigcup_{i=0}^n C_i \mid \forall b \in B \setminus \bigcup_{i=0}^n D_i, g(b) \neq a\},$$

and we can prove that $C_{n+1} = \{g(b) \mid b \in D_n\}$, $D_{n+1} = \{f(a) \mid a \in C_n\}$.

Define

$$h(a) = \begin{cases} f(a) & \text{if } a \in \bigcup_{n=0}^{\infty} C_n, \\ \text{the unique } b \text{ such that } g(b) = a & \text{if } a \notin \bigcup_{n=0}^{\infty} C_n. \end{cases}$$

(Note: For $a \notin \bigcup C_n$, a is in the image of g , and since g is injective, the b is unique.)

Prove h is a bijection.

Injection: Suppose $h(a_1) = h(a_2)$. Cases:

- $a_1 \in C_n, a_2 \in C_m$: then $h(a_1) = f(a_1) = f(a_2) = h(a_2) \Rightarrow a_1 = a_2$ (since f injective).
- $a_1, a_2 \notin \bigcup C_i$: then $a_1 = g(h(a_1)) = g(h(a_2)) = a_2$.
- $a_1 \notin \bigcup C_i, a_2 \in C_n$ for some n : then $f(a_2) \in D_n$ and $g(f(a_2)) \in C_{n+1}$.

Thus $a_1 = g(h(a_1)) = g(h(a_2)) = g(f(a_2)) \in C_{n+1}$ **Contradiction!**

Surjection: Cases:

- Suppose $b \in \bigcup_{n=0}^{\infty} D_n$. Obvious.
- Suppose $b \notin \bigcup_{n=0}^{\infty} D_n$. Let $a = g(b)$.

Then $a \in A \setminus \bigcup_{n=0}^{\infty} C_n$ otherwise $a \in C_n$, then $b \in D_n$, **Contradiction!**

1.8 Equinumerosity

Definition 1.14

- $A \preceq B$ if there exists an injection $A \rightarrow B$.
- $A \approx B$ ($A \cong B$) if there exists a bijection $A \rightarrow B$.



Theorem 1.19

$[0, 1] \approx [0, 1)$.



Proof By Bernstein's theorem, or explicitly: $f(x) = x/2$ for $x = 1, 1/2, 1/4, \dots$ and $f(x) = x$ otherwise gives a bijection $[0, 1] \rightarrow [0, 1)$.

Theorem 1.20

$\mathbb{N} \approx \mathbb{N} \times \mathbb{N}$. $\mathbb{Q} \preceq \mathbb{Z}^+ \times \mathbb{Z} \approx \mathbb{N} \times \mathbb{N} \approx \mathbb{N}$.



Theorem 1.21

If $A_1 \approx B_1$ and $A_2 \approx B_2$, then $(A_1 \rightarrow A_2) \approx (B_1 \rightarrow B_2)$.



Proof Suppose $f : A_1 \rightarrow B_1$ and $g : A_2 \rightarrow B_2$ are bijections. Define $H(h) = g \circ h \circ f^{-1}$ for $h : A_1 \rightarrow A_2$. Then H is a bijection from $(A_1 \rightarrow A_2)$ to $(B_1 \rightarrow B_2)$.

Theorem 1.22

If $A_1 \preceq B_1$ and $A_2 \preceq B_2$, then $(A_1 \rightarrow A_2) \preceq (B_1 \rightarrow B_2)$.



Proof Let $f : A_1 \rightarrow B_1$ and $g : A_2 \rightarrow B_2$ be injections.

If $B_2 = \emptyset$, then $A_2 = \emptyset$, and both function sets are empty, so trivial.

Assume $B_2 \neq \emptyset$, pick $b_0 \in B_2$. For any $h : A_1 \rightarrow A_2$, define $H(h) : B_1 \rightarrow B_2$ by

$$H(h)(b) = \begin{cases} g(h(a)) & \text{if } \exists a \in A_1 \text{ with } f(a) = b, \\ b_0 & \text{otherwise.} \end{cases}$$

Since f is injective, the a is unique.

Then H is injective: if $H(h_1) = H(h_2)$, then for any $a \in A_1$, $H(h_1)(f(a)) = H(h_2)(f(a))$, so $g(h_1(a)) = g(h_2(a))$, thus $h_1(a) = h_2(a)$ because g injective. Hence $h_1 = h_2$.

Theorem 1.23

$(A \times B) \rightarrow C \approx A \rightarrow (B \rightarrow C)$.



Proof For $f : A \times B \rightarrow C$, define $H(f) : A \rightarrow (B \rightarrow C)$ by $H(f)(a)(b) = f(a, b)$. This is a bijection.

- **Injection:** Suppose $H(f_1) = H(f_2)$, $a \in A, b \in B$, then $H(f_1)(a)(b) = H(f_2)(a)(b)$. Since a, b are arbitrary, $f_1 = f_2$.
- **Surjection:** Suppose $g : A \rightarrow (B \rightarrow C)$, let $f(a, b) = g(a)(b)$, thus $H(f)(a)(b) = f(a, b) = g(a)(b)$. So $H(f) = g$.

Theorem 1.24

$$A \rightarrow (B \times C) \approx (A \rightarrow B) \times (A \rightarrow C).$$



Proof Define $H(f) = (H_1(f), H_2(f))$ where $H_1(f)(a) = \text{fst}(f(a))$, $H_2(f)(a) = \text{snd}(f(a))$. This is a bijection.

- Injection: if $H(f) = H(f')$, then for all a , $f(a) = (H_1(f)(a), H_2(f)(a)) = (H_1(f')(a), H_2(f')(a)) = f'(a)$.
- Surjection: given $g : A \rightarrow B$, $h : A \rightarrow C$, define $f(a) = (g(a), h(a))$, then $H(f) = (g, h)$.

We can also use another method:

Proof Let $H(f, g) = (f(a), g(a))$, $\forall f : A \rightarrow B$, $\forall g : A \rightarrow C$. This is a bijection.

- Injection: Suppose $H(f_1, g_1) = H(f_2, g_2)$, then $\forall a$, $H(f_1, g_1)(a) = H(f_2, g_2)(a)$, i.e. $(f_1(a), g_1(a)) = (f_2(a), g_2(a))$. Thus $f_1 = f_2$, $g_1 = g_2$.
- Surjection: Consider $h : A \rightarrow B \times C$. Let $f(a) = \text{fst}(h(a))$, $g(a) = \text{snd}(h(a))$, then $\forall a$, $H(f, g)(a) = (f(a), g(a)) = h(a)$, so $H(f, g) = h$.

Theorem 1.25

$$(\{0, 1\} \rightarrow A) \approx A \times A.$$



Proof Define $H(f) = (f(0), f(1))$. This is a bijection.

- Injection: if $H(f_1) = H(f_2)$, then $f_1(0) = f_2(0)$, $f_1(1) = f_2(1)$. So $f_1 = f_2$.
- Surjection: Suppose $a, b \in A$, let

$$f(x) = \begin{cases} a & x = 0, \\ b & x = 1. \end{cases}$$

Then $H(f) = (a, b)$.

Definition 1.15

$$A \rightarrow B \text{ is denoted } B^A.$$

$$\text{Then } C^{A \times B} = (C^B)^A, (B \times C)^A = B^A \times C^A, A^2 = A \times A.$$

**Theorem 1.26**

$$\text{If } A \approx B, \text{ then } \mathcal{P}(A) \approx \mathcal{P}(B).$$



Proof Let $f : A \rightarrow B$ be a bijection. Define $H : \mathcal{P}(A) \rightarrow \mathcal{P}(B)$ by $H(X) = \{f(a) \mid a \in X\}$. H is a bijection.

- Injection: if $H(X) = H(Y)$, then for any $a \in X$, $f(a) \in H(Y)$, so $\exists b \in Y$ with $f(a) = f(b)$, hence $a = b \in Y$, so $X \subseteq Y$, similarly $Y \subseteq X$.
- Surjection: for $Y \subseteq B$, let $X = \{f^{-1}(b) \mid b \in Y\}$, then $H(X) = Y$.

Theorem 1.27

$$(A \rightarrow \{0, 1\}) \approx \mathcal{P}(A), \text{ i.e., } 2^A \approx \mathcal{P}(A).$$



Proof Define $H(f) = \{a \in A \mid f(a) = 1\}$. This is a bijection.

- Injection: if $H(f) = H(g)$, then $f(a) = 1 \iff a \in H(f) \iff a \in H(g) \iff g(a) = 1$, and since 0, 1 are the only values, $f(a) = 0$ iff $g(a) = 0$, so $f = g$.
- Surjection: for $X \subseteq A$, define $f(a) = 1$ if $a \in X$, else 0, then $H(f) = X$.

1.9 Countable Sets

Definition 1.16

A set A is called **countable** if $A \approx \mathbb{N}$ (i.e., there exists a bijection between A and the natural numbers). 

Example 1.5 The following sets are countable:

- $\mathbb{N}$ itself,
- $\mathbb{Z}^+$,
- $\mathbb{Z}$,
- $\{n \in \mathbb{N} \mid n \text{ is even}\}$,
- $\mathbb{N} \times \mathbb{N}, \mathbb{N} \times \mathbb{N} \times \mathbb{N}$,
- $\mathbb{Q}$.

Proof There exists a bijection: $f : x \rightarrow x + 1$ is from $\mathbb{N}$ to $\mathbb{Z}^+$.

There exists a bijection: $f : (i, j) \rightarrow i + \frac{1}{2}(1 + i + j)(i + j)$ is from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$.

There exists a bijection: $f : \frac{p}{q} \rightarrow (p, q)$ is from $\mathbb{Q}$ to $\mathbb{N} \times \mathbb{Z}^+$.

Theorem 1.28

The set of all finite sequences of natural numbers is countable. 

Corollary 1.1

The following sets are countable:

- The set of strings (of finite length) over a countable alphabet,
 - The set of integer arithmetic expressions,
 - The set of polynomials with integer coefficients,
 - The set of algebraic numbers (roots of polynomials with integer coefficients).
- 

1.10 Uncountable Sets

Theorem 1.29 (Cantor)

For any set X , there is no bijection between X and $\mathcal{P}(X)$; in particular, $X \not\approx \mathcal{P}(X)$. 

Proof Suppose that there exists a bijection $\theta : X \rightarrow \mathcal{P}(X)$.

Define the set

$$Y = \{x \in X \mid x \notin \theta(x)\}.$$

Clearly $Y \subseteq X$, hence $Y \in \mathcal{P}(X)$. Since θ is surjective, there exists some $y \in X$ such that $\theta(y) = Y$.

Now we examine the membership of y in Y :

- If $y \in Y$, then by the definition of Y , we have $y \notin \theta(y)$. But $\theta(y) = Y$, so $y \notin Y$, contradiction!
- If $y \notin Y$, then since $\theta(y) = Y$, we have $y \notin \theta(y)$. By the definition of Y , this precisely implies $y \in Y$, contradiction!

Hence $X \not\approx \mathcal{P}(X)$.

Corollary 1.2

$\mathcal{P}(\mathbb{N})$ is uncountable; consequently, $(\mathbb{N} \rightarrow \{0, 1\})$ is uncountable.



Theorem 1.30

The set of infinite sequences of natural numbers is uncountable.



Proof Every infinite sequence of natural numbers is a function from $\mathbb{N}$ to $\mathbb{N}$. Since

$$2^{\mathbb{N}} \preceq \mathbb{N}^{\mathbb{N}} \preceq (2^{\mathbb{N}})^{\mathbb{N}} \approx 2^{\mathbb{N} \times \mathbb{N}} \approx 2^{\mathbb{N}},$$

we have $\mathbb{N}^{\mathbb{N}} \approx 2^{\mathbb{N}}$, which is uncountable.

Theorem 1.31

$\mathbb{R} \approx 2^{\mathbb{N}}$.



Proof It suffices to show $(0, 1) \approx 2^{\mathbb{N}}$. ($\mathbb{R} \approx (0, 1)$ via, e.g., $x \mapsto \tan(\pi(x - \frac{1}{2}))$)

First, define an injection $g : (0, 1) \rightarrow 2^{\mathbb{N}}$ as follows: for each $x \in (0, 1)$, write its binary expansion uniquely as $x = 0.b_1b_2b_3 \dots$ with the convention that the expansion does not end in all 1's (i.e., choose the terminating expansion when possible). Then $g(x) = (b_1, b_2, \dots)$ is well-defined and injective, because different binary expansions chosen this way represent different real numbers.

Second, define an injection $h : 2^{\mathbb{N}} \rightarrow (0, 1)$ by $h((a_n)) = 0.(a_1 + 1)(a_2 + 1) \dots$ interpreted in base 10 (or base 3), where each digit is 1 or 2. Since the decimal representation using only digits 1 and 2 is unique (no ambiguity with trailing 9's), h is injective.

Thus by Bernstein, $(0, 1) \approx 2^{\mathbb{N}}$.

Corollary 1.3

$(0, 1) \approx [0, 1] \approx 2^{\mathbb{N}}$.



Theorem 1.32

$(\mathbb{R} \rightarrow \mathbb{R}) \approx 2^{2^{\mathbb{N}}}$.



Proof We have $\mathbb{R}^{\mathbb{R}} \approx (2^{\mathbb{N}})^{2^{\mathbb{N}}} \approx 2^{\mathbb{N} \times 2^{\mathbb{N}}}$. Moreover,

$$2^{2^{\mathbb{N}}} \preceq 2^{\mathbb{N} \times 2^{\mathbb{N}}} \preceq 2^{2^{\mathbb{N}} \times 2^{\mathbb{N}}} \approx 2^{(2^{\mathbb{N}})^2} \preceq 2^{(2^{\mathbb{N}})^{\mathbb{N}}} \approx 2^{2^{\mathbb{N}} \times \mathbb{N}} \approx 2^{2^{\mathbb{N}}}.$$

Thus all inequalities are equalities, so $(\mathbb{R} \rightarrow \mathbb{R}) \approx 2^{2^{\mathbb{N}}}$.

Theorem 1.33

The set of continuous functions from $\mathbb{R}$ to $\mathbb{R}$ is equinumerous with $\mathbb{R}$.



Proof A continuous function $F : \mathbb{R} \rightarrow \mathbb{R}$ can be determined by its values on $\mathbb{Q}$.

If $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are continuous and $f(q) = g(q)$ for all $q \in \mathbb{Q}$, then $f = g$.

because for any $x \in \mathbb{R}$, we can find a sequence of rationals x_n converging to x , so $f(x) = \lim f(x_n) = \lim g(x_n) = g(x)$.

Example 1.6 Let A be the set of strictly increasing sequences of natural numbers(infinite length). Specifically, a element of A is $a_1 a_2 a_3 \dots$ s.t. $\forall n \in \mathbb{N}, a_n \in \mathbb{N}$ and $a_n < a_{n+1}$.

a) Prove A is uncountable.

b) For $f : \mathbb{N} \rightarrow A$, find $a \in A$ such that $\forall n \in \mathbb{N}, f(n) \neq a$.

Proof a) We can define an injection from $2^{\mathbb{N}}$ to A : for $2^{\mathbb{N}}$ which is a infinite sequence of 0's and 1's, we can add the position of each 1 to the string to get a strictly increasing sequence of natural numbers. Hence A is uncountable.

b) Let A be the set of strictly increasing sequences of natural numbers.

Given $f : \mathbb{N} \rightarrow A$, construct $a = (a_n)$ recursively by

$$a_1 = f(1)_1 + 1, \quad a_{n+1} = \begin{cases} a_n + 1, & f(n)_{n+1} - f(n)_n > 1, \\ a_n + 2, & f(n)_{n+1} - f(n)_n = 1. \end{cases}$$

Since $a_{n+1} > a_n$, a is strictly increasing, so $a \in A$.

Given a fixed k .

- If $a_k \neq f(k)_k$, done.
- Otherwise $a_k = f(k)_k$. Let $d = f(k)_{k+1} - f(k)_k \geq 1$. By the rule, $a_{k+1} = f(k)_k + 1$ if $d > 1$, or $f(k)_k + 2$ if $d = 1$; in either case $a_{k+1} \neq f(k)_{k+1}$. Hence $a \neq f(k)$.

Since k was arbitrary, we have $a \neq f(k)$ for all $k \in \mathbb{N}$.